

Evolution of Intel Microprocessors

Key Differences: 8086 to 80486

1 Key Differences Among Intel 8086 to 80486

Feature	8086	80186	80286	80386	80486
Processor Architecture	16-bit	16-bit	16-bit	32-bit	32-bit
Data Bus Width	16-bit	16-bit	16-bit	32-bit	32-bit
Address Bus Width	20-bit	20-bit	24-bit	32-bit	32-bit
Maximum Physical Memory	1 MB	1 MB	16 MB	4 GB	4 GB
Operating Modes	Real	Real	Real, Protected	Real, Protected, Virtual	Real, Protected, Virtual
Protected Mode	Not supported	Not supported	Supported	Supported	Supported
Paging Support	Not supported	Not supported	Not supported	Supported	Supported
On-Chip Peripherals	None	Integrated peripherals	None	None	Cache and FPU
Cache Memory	Not available	Not available	Not available	Not available	8 KB on-chip
Floating Point Unit	External (8087)	External	External	External (80387)	Built-in
Clock Generator	External	On-chip	External	External	External
Multitasking Support	Not supported	Not supported	Supported	Supported	Supported
Primary Application	General computing	Embedded systems	Multi-user systems	Advanced operating systems	High-performance systems

2 Summary of Evolution

- **8086:** First 16-bit x86 processor with basic real-mode operation.
- **80186:** Enhanced 8086 with integrated peripherals, mainly for embedded systems.
- **80286:** Introduced protected mode and expanded memory addressing.
- **80386:** First true 32-bit processor with paging and virtual memory.
- **80486:** Integrated cache and floating-point unit with pipelining for high performance.